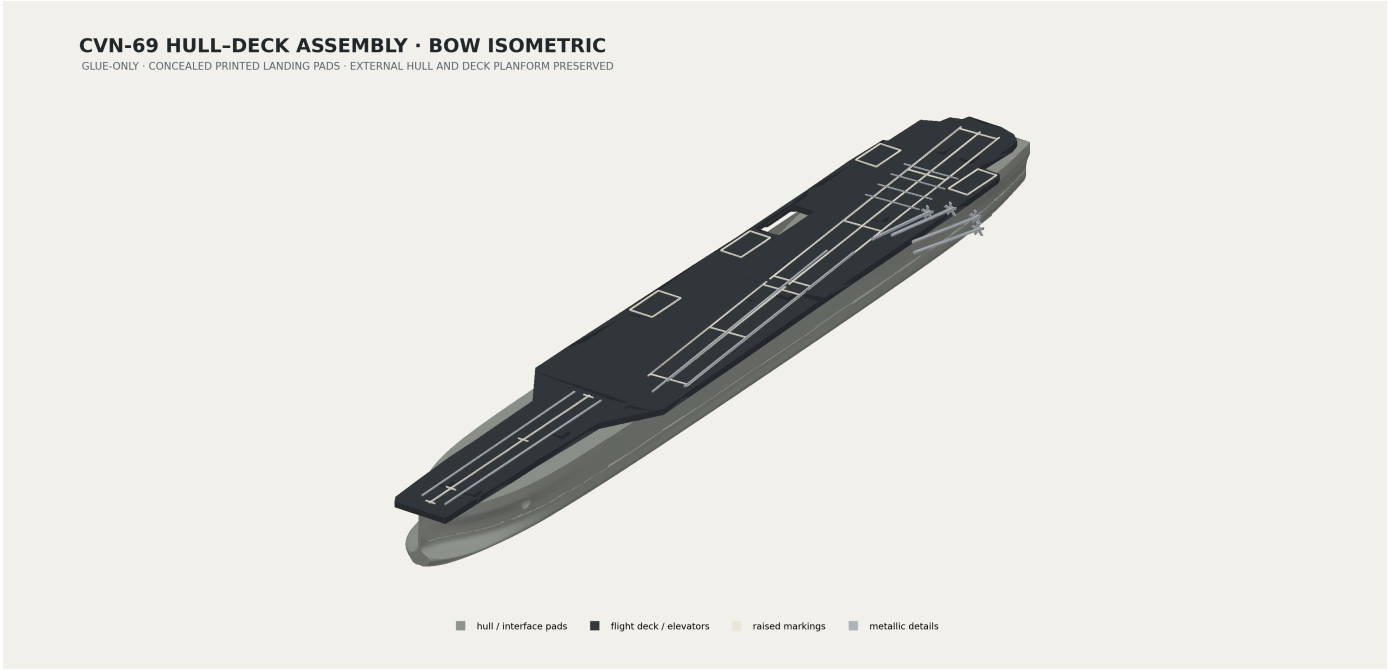


# CVN-69 Milestone 2 Printing Guide

BAMBU-READY · 0.4 MM NOZZLE · GLUE-ONLY · 1:700



## Print files

| 3MF                        | Contents                               | Validated size (mm)     |
|----------------------------|----------------------------------------|-------------------------|
| Print_Plate_01_Hull.3mf    | 3 hull modules + approved running gear | 220.06 × 177.29 × 31.43 |
| Print_Plate_02_Deck.3mf    | 3 socketed deck modules                | 197.00 × 217.95 × 3.00  |
| Print_Plate_03_Details.3mf | 19 deck details + 12 interface pads    | 216.80 × 171.59 × 2.40  |
| Interface_Test_Coupon.3mf  | male/female clearance test             | 53.00 × 20.00 × 5.20    |

## Orientation and support

- Hull modules print with their flat hull-top datum downward. Interface sockets are open to the bed; maximum socket roof span is 6.50 mm.
- Deck modules print underside down. The new sockets are open and shallow; visible deck surfaces and planform are unchanged.
- Elevators, markings, tracks, wires, and landing pads print flat. Use a brim only if your build surface needs it for the longest thin details.
- Do not auto-scale. Every production STL and print 3MF is already millimetres at 1:700.

# Material mapping by object name

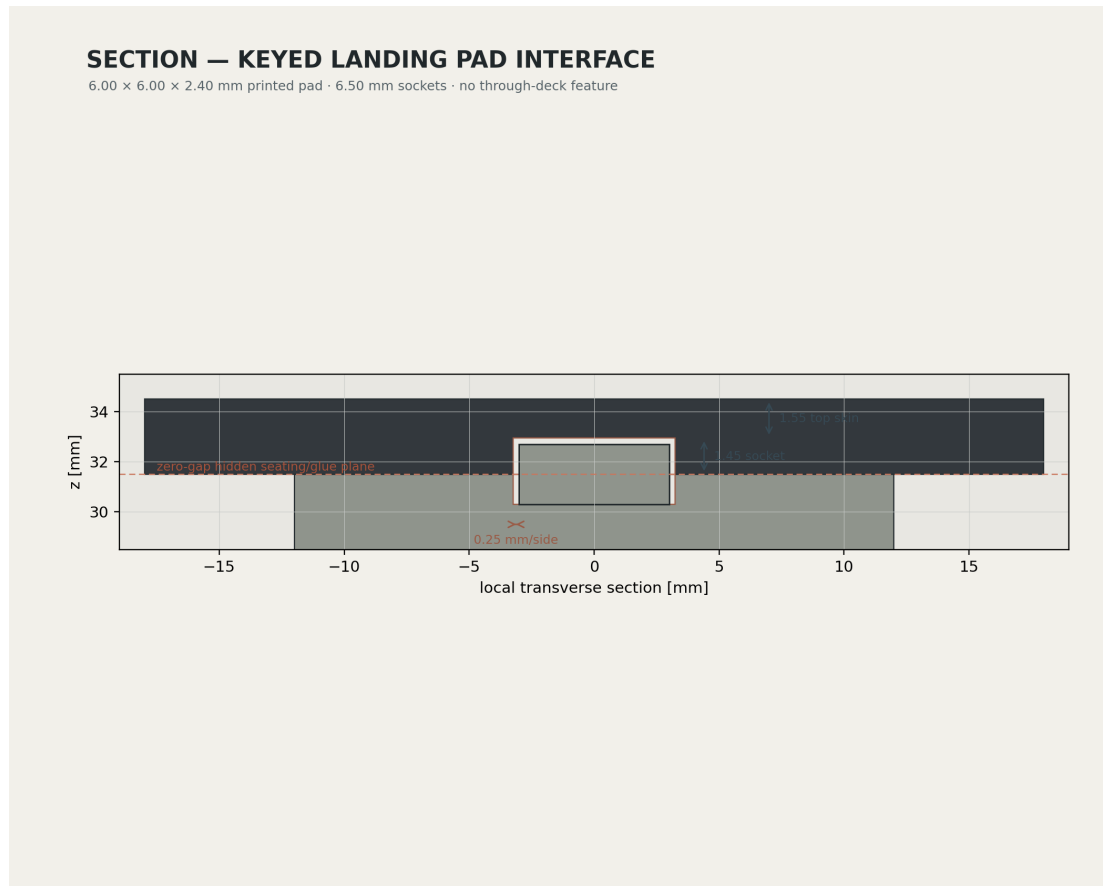
This table is also supplied as Material\_Mapping.csv. AMS slot numbers are deliberately not assigned.

| Object name                         | Material                    |
|-------------------------------------|-----------------------------|
| Arresting_Wire_1                    | Bambu PLA Silk Silver       |
| Arresting_Wire_2                    | Bambu PLA Silk Silver       |
| Arresting_Wire_3                    | Bambu PLA Silk Silver       |
| Arresting_Wire_4                    | Bambu PLA Silk Silver       |
| Catapult_1_Bow_Port                 | Bambu PLA Silk Silver       |
| Catapult_2_Bow_Starboard            | Bambu PLA Silk Silver       |
| Catapult_3_Waist_Inner              | Bambu PLA Silk Silver       |
| Catapult_4_Waist_Outer              | Bambu PLA Silk Silver       |
| Deck_Module_1_Bow                   | Bambu PLA Matte Charcoal    |
| Deck_Module_2_Midship               | Bambu PLA Matte Charcoal    |
| Deck_Module_3_Stern                 | Bambu PLA Matte Charcoal    |
| Elevator_1_Port                     | Bambu PLA Matte Charcoal    |
| Elevator_2_Starboard                | Bambu PLA Matte Charcoal    |
| Elevator_3_Starboard                | Bambu PLA Matte Charcoal    |
| Elevator_4_Starboard                | Bambu PLA Matte Charcoal    |
| Hull_Module_1_Bow                   | Bambu PLA Matte Ash Gray    |
| Hull_Module_2_Midship               | Bambu PLA Matte Ash Gray    |
| Hull_Module_3_Stern                 | Bambu PLA Matte Ash Gray    |
| Interface_Pad_01_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_01_Starboard          | Bambu PLA Matte Ash Gray    |
| Interface_Pad_02_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_02_Starboard          | Bambu PLA Matte Ash Gray    |
| Interface_Pad_03_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_03_Starboard          | Bambu PLA Matte Ash Gray    |
| Interface_Pad_04_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_04_Starboard          | Bambu PLA Matte Ash Gray    |
| Interface_Pad_05_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_05_Starboard          | Bambu PLA Matte Ash Gray    |
| Interface_Pad_06_Port               | Bambu PLA Matte Ash Gray    |
| Interface_Pad_06_Starboard          | Bambu PLA Matte Ash Gray    |
| Propeller_1                         | Bambu PLA Silk Silver       |
| Propeller_2                         | Bambu PLA Silk Silver       |
| Propeller_3                         | Bambu PLA Silk Silver       |
| Propeller_4                         | Bambu PLA Silk Silver       |
| Raised_Marking_Bow_Centerline       | Bambu PLA Matte Ivory White |
| Raised_Marking_Elevator_1_Port      | Bambu PLA Matte Ivory White |
| Raised_Marking_Elevator_2_Starboard | Bambu PLA Matte Ivory White |
| Raised_Marking_Elevator_3_Starboard | Bambu PLA Matte Ivory White |
| Raised_Marking_Elevator_4_Starboard | Bambu PLA Matte Ivory White |
| Raised_Marking_Landing_Aft          | Bambu PLA Matte Ivory White |
| Raised_Marking_Landing_Forward      | Bambu PLA Matte Ivory White |
| Rudder_Port                         | Bambu PLA Matte Ash Gray    |
| Rudder_Starboard                    | Bambu PLA Matte Ash Gray    |

| Object name     | Material                 |
|-----------------|--------------------------|
| Shaft_1         | Bambu PLA Silk Silver    |
| Shaft_1_Strut_P | Bambu PLA Matte Ash Gray |
| Shaft_1_Strut_S | Bambu PLA Matte Ash Gray |
| Shaft_2         | Bambu PLA Silk Silver    |
| Shaft_2_Strut_P | Bambu PLA Matte Ash Gray |
| Shaft_2_Strut_S | Bambu PLA Matte Ash Gray |
| Shaft_3         | Bambu PLA Silk Silver    |
| Shaft_3_Strut_P | Bambu PLA Matte Ash Gray |
| Shaft_3_Strut_S | Bambu PLA Matte Ash Gray |
| Shaft_4         | Bambu PLA Silk Silver    |
| Shaft_4_Strut_P | Bambu PLA Matte Ash Gray |
| Shaft_4_Strut_S | Bambu PLA Matte Ash Gray |

# Interface fit and assembly readiness

Print and test the coupon before the complete assembly. The designed lateral clearance is 0.25 mm per side; the deck should seat flat before adhesive is applied.



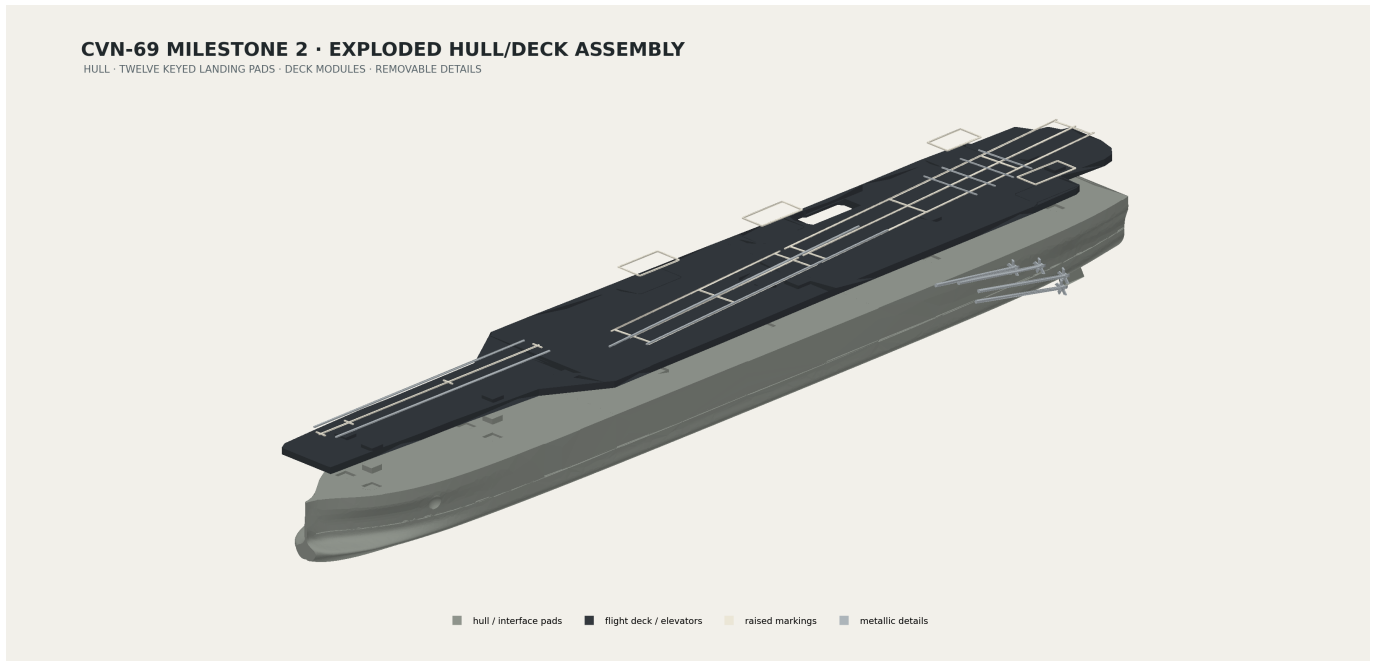
## Acceptance checks

- Male feature enters without force and has no elephant-foot interference.
- Female face seats against the male base with no visible rocking or gap over 0.30 mm.
- A small amount of the intended adhesive cures without locking the parts before full seating.
- If the coupon fails, adjust slicer compensation or reprint; do not scale the production model.

# Glue-only assembly sequence

Use the complete procedure in Assembly/Glue\_Only\_Assembly.md. The short sequence is:

- Assemble and cure the approved hull modules on a straight datum.
- Assemble and cure the transformed deck modules on a flat datum.
- Dry-fit all twelve pads in the hull, then lower the deck vertically and verify full seating.
- Apply a thin adhesive film only to hidden seating areas and sockets; reinstall without longitudinal sliding.
- Install elevators and raised details after the hull/deck bond has cured.



Physical qualification is complete: the 0.25 mm-per-side coupon assembled by hand and seated correctly at 100% scale with a 0.40 mm nozzle, 0.16 mm layers, three walls, 0.00 mm XY compensation, and 0.15 mm elephant-foot compensation. Production interface dimensions are frozen.